Advanced Drone Technology – Introduction to Guidance and Control Using Python Simulations



INDIA SPACE LAB

Project Report

17th June to 31st July 2026

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Aim

The aim of this project is to understand the principles of guidance and control systems through Python-based simulations. The project focuses on implementing and tuning a PID (Proportional–Integral–Derivative) controller for drone altitude control and studying autonomous boat guidance under different environmental conditions.

Objectives

- To understand the working principle of guidance and control systems.
- To implement and tune a PID controller for drone altitude control.
- To analyze the effect of proportional, integral, and derivative gains on system performance.
- To study the influence of external disturbances on autonomous systems.
- To understand autonomous boat guidance and path-following algorithms.
- To compare the performance of the controller with and without disturbances.
- To gain practical experience using Python simulations in Google Colab.

Introduction

Guidance and control systems are essential components of modern autonomous vehicles such as drones, aircraft, satellites, rockets, and autonomous boats. These systems ensure that a vehicle follows a desired path while maintaining stability and accuracy under varying operating conditions. Feedback control techniques are widely used to continuously monitor system performance and automatically correct deviations from the desired trajectory.

This project demonstrates the application of guidance and control concepts using Python simulations in Google Colab. Two practical simulations were performed. The first simulation involved controlling the altitude of a drone using a PID controller, while the second simulation demonstrated autonomous boat guidance and path tracking. Different controller gains were adjusted to study their effect on system response and disturbance rejection.

Theory

Guidance and Control Systems

A guidance system determines the desired path of an autonomous vehicle, while the control system ensures that the vehicle accurately follows the planned trajectory. Together, these systems enable autonomous navigation, stability, and precise motion control.

PID Controller

A PID (Proportional–Integral–Derivative) controller is one of the most widely used feedback controllers in engineering. The proportional component reacts to the current error, the integral component removes accumulated steady-state error, and the derivative component predicts future error based on the rate of change. Proper tuning of these three parameters improves system stability, reduces overshoot, minimizes oscillations, and provides faster settling time.

Autonomous Boat Guidance

Autonomous boat guidance uses feedback algorithms to follow a predefined path while compensating for environmental disturbances such as water currents. Controller tuning minimizes path-tracking errors and improves navigation accuracy.

Environmental Disturbances

External disturbances such as wind and water currents affect the stability of autonomous systems. A properly tuned controller enables the system to reject disturbances and return to the desired operating condition with minimum error.

Software Used

- Google Colab
- Python
- NumPy
- Matplotlib

Methodology

Task 1 – Drone Altitude Control

The provided Google Colab notebook was executed to simulate drone altitude control using a PID controller. The proportional (P), integral (I), and derivative (D) gains were adjusted to obtain stable altitude control. Wind disturbance was introduced during the simulation to evaluate the controller's ability to reject disturbances and maintain stability. Different parameter combinations were tested to achieve minimum overshoot, reduced oscillations, and faster settling time.

Task 2 – Autonomous Boat Guidance

The autonomous boat guidance simulation was executed to analyze trajectory tracking performance. Controller parameters were adjusted while introducing environmental disturbances to

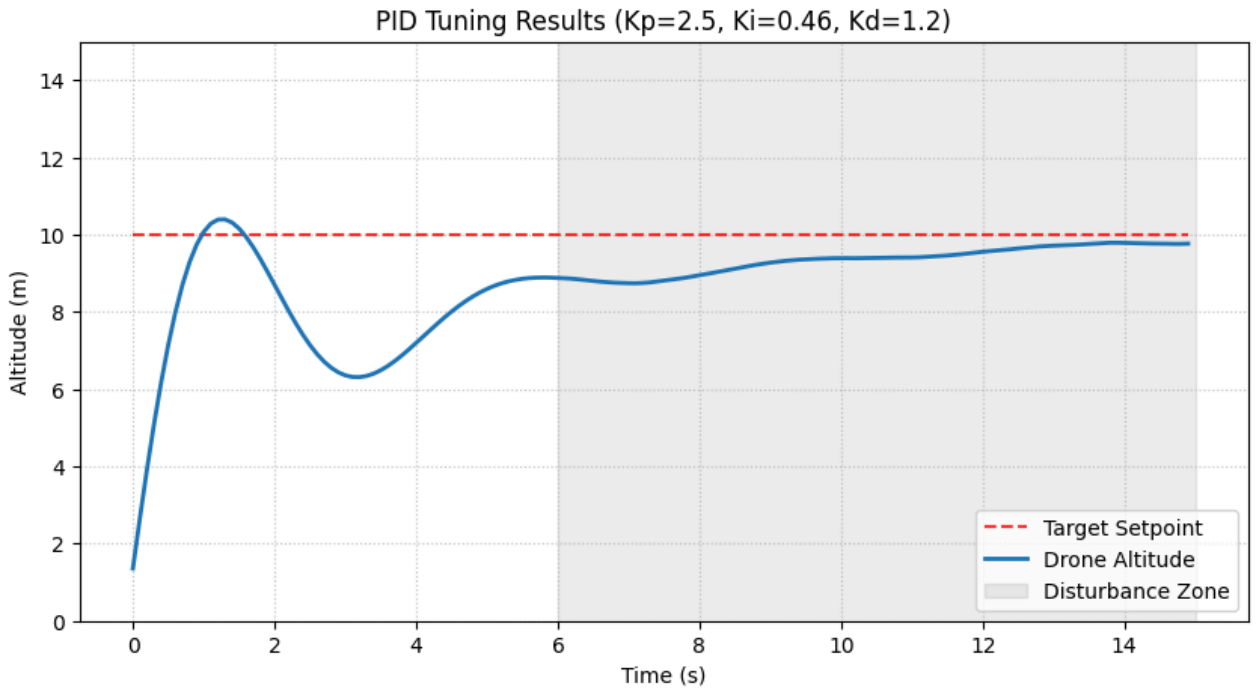
observe their effect on path-following accuracy. Simulations with and without disturbances were compared to evaluate system stability and navigation performance.

Bonus Task

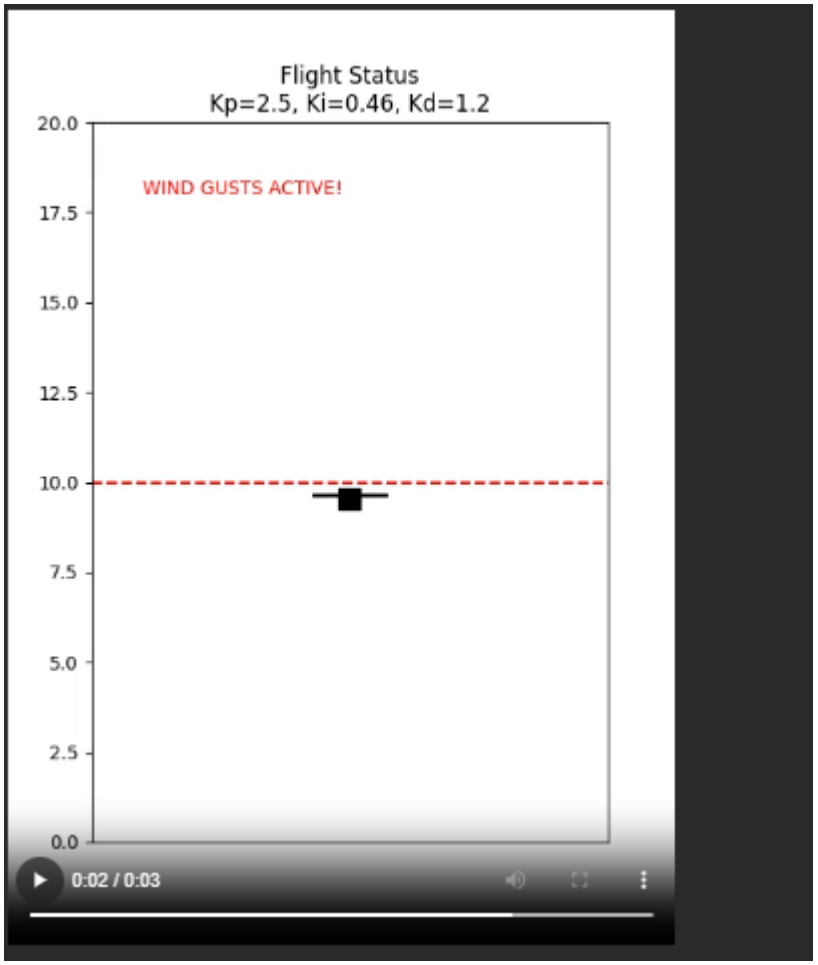
The optional figure-eight trajectory simulation was executed to demonstrate waypoint tracking and continuous path-following capabilities of the autonomous system.

Results

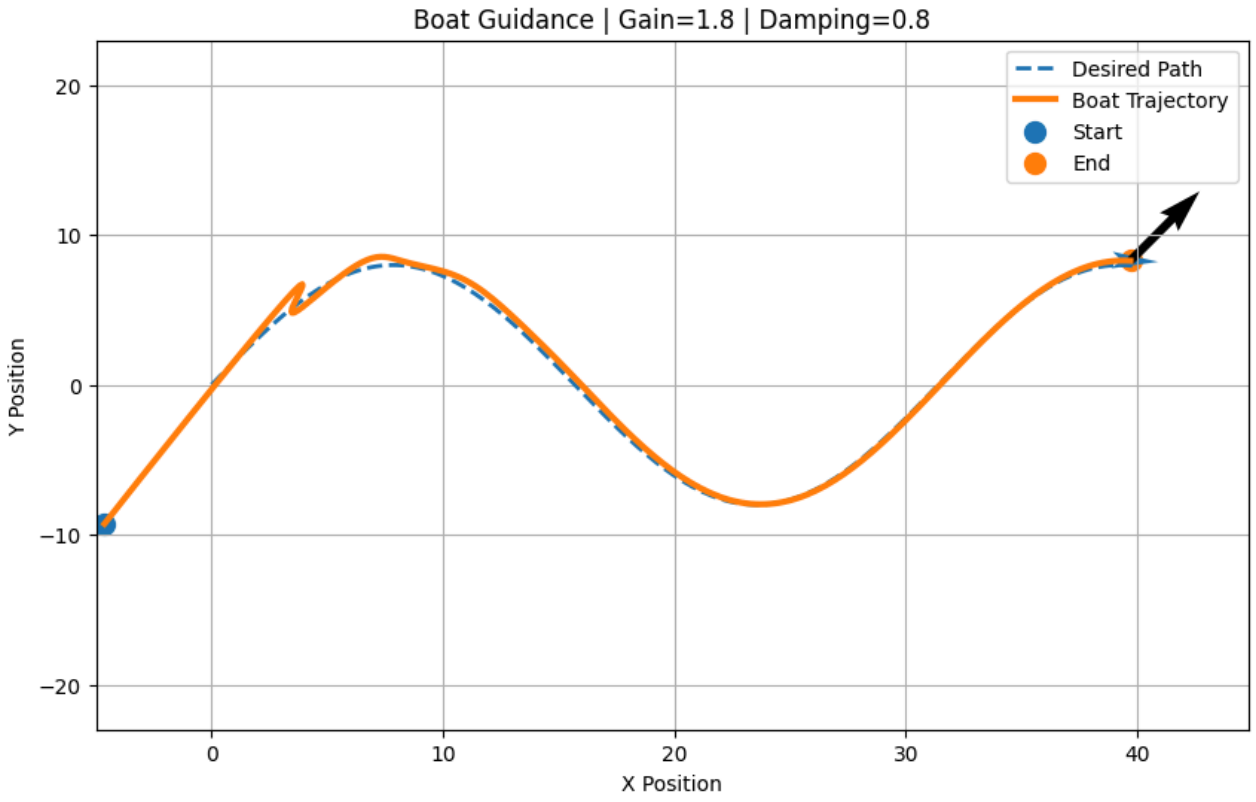
- PID Controller Tuning Graph



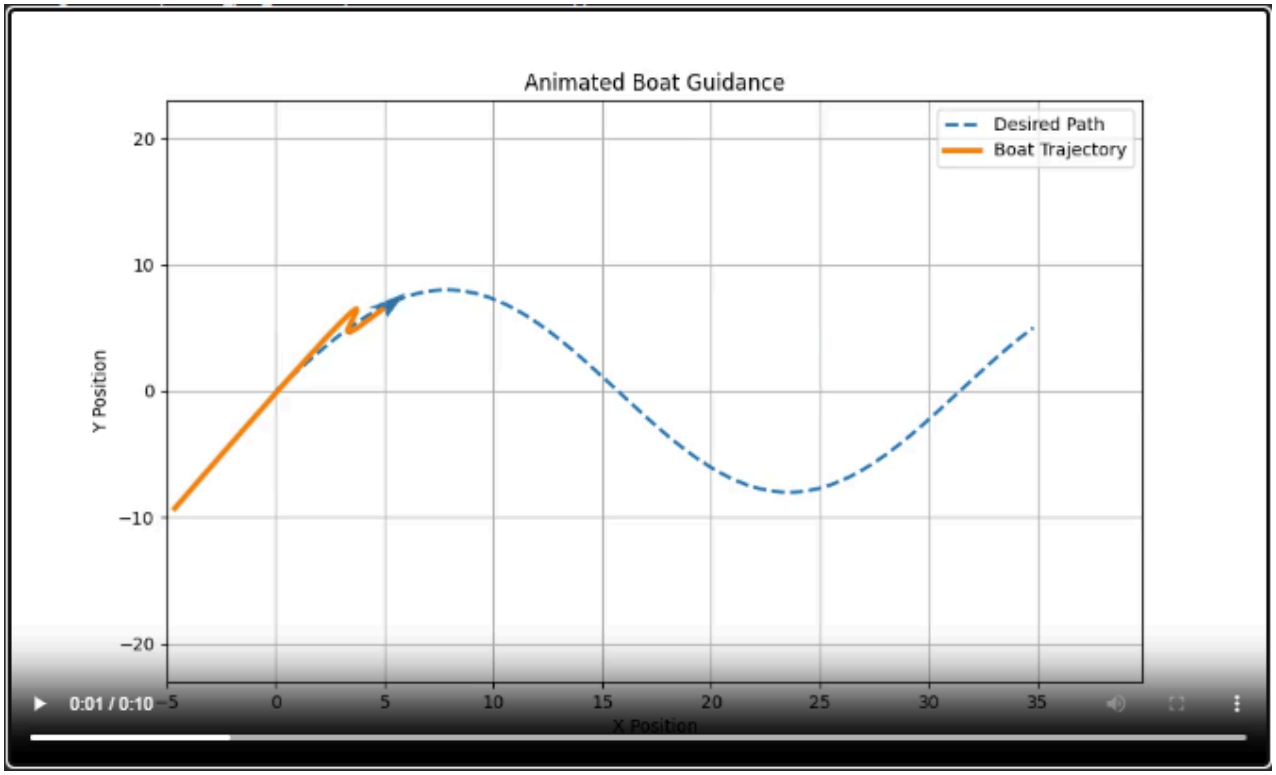
- Drone Altitude Simulation



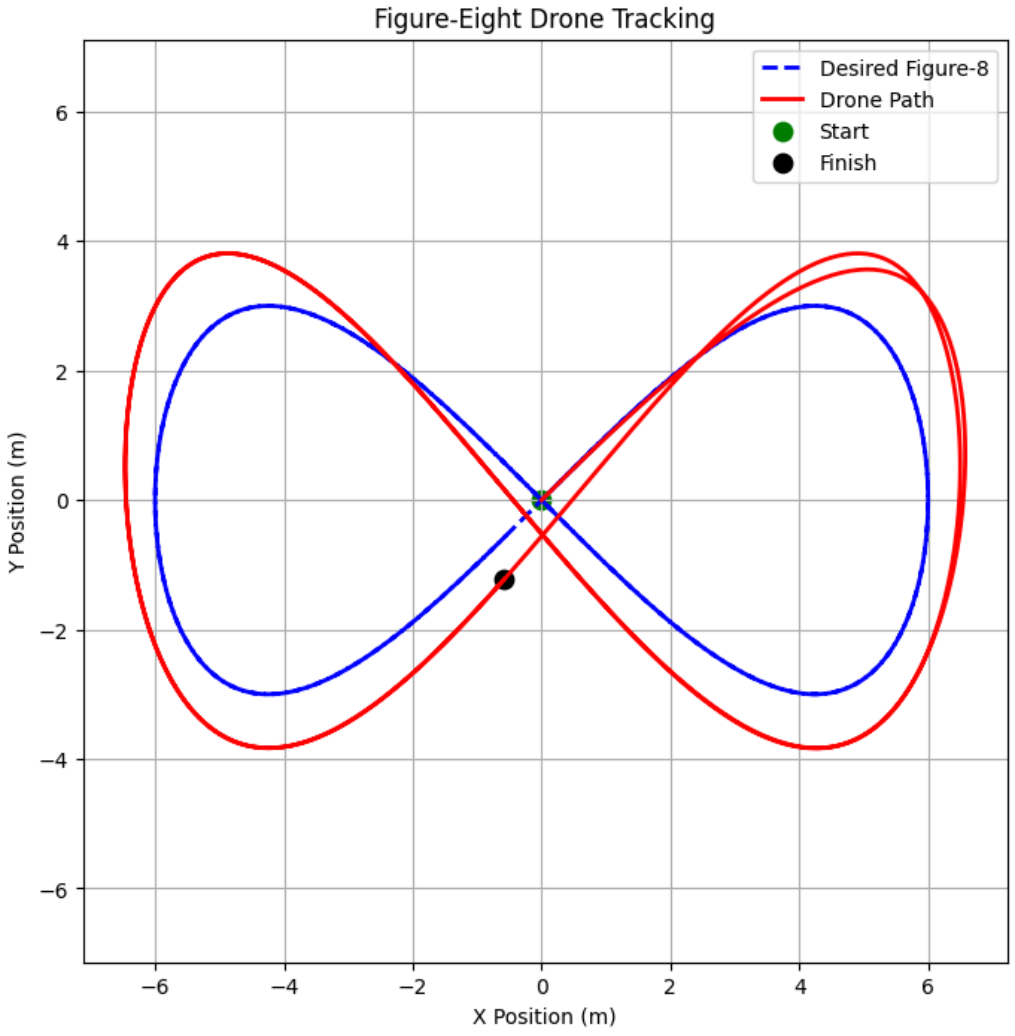
- Boat Guidance

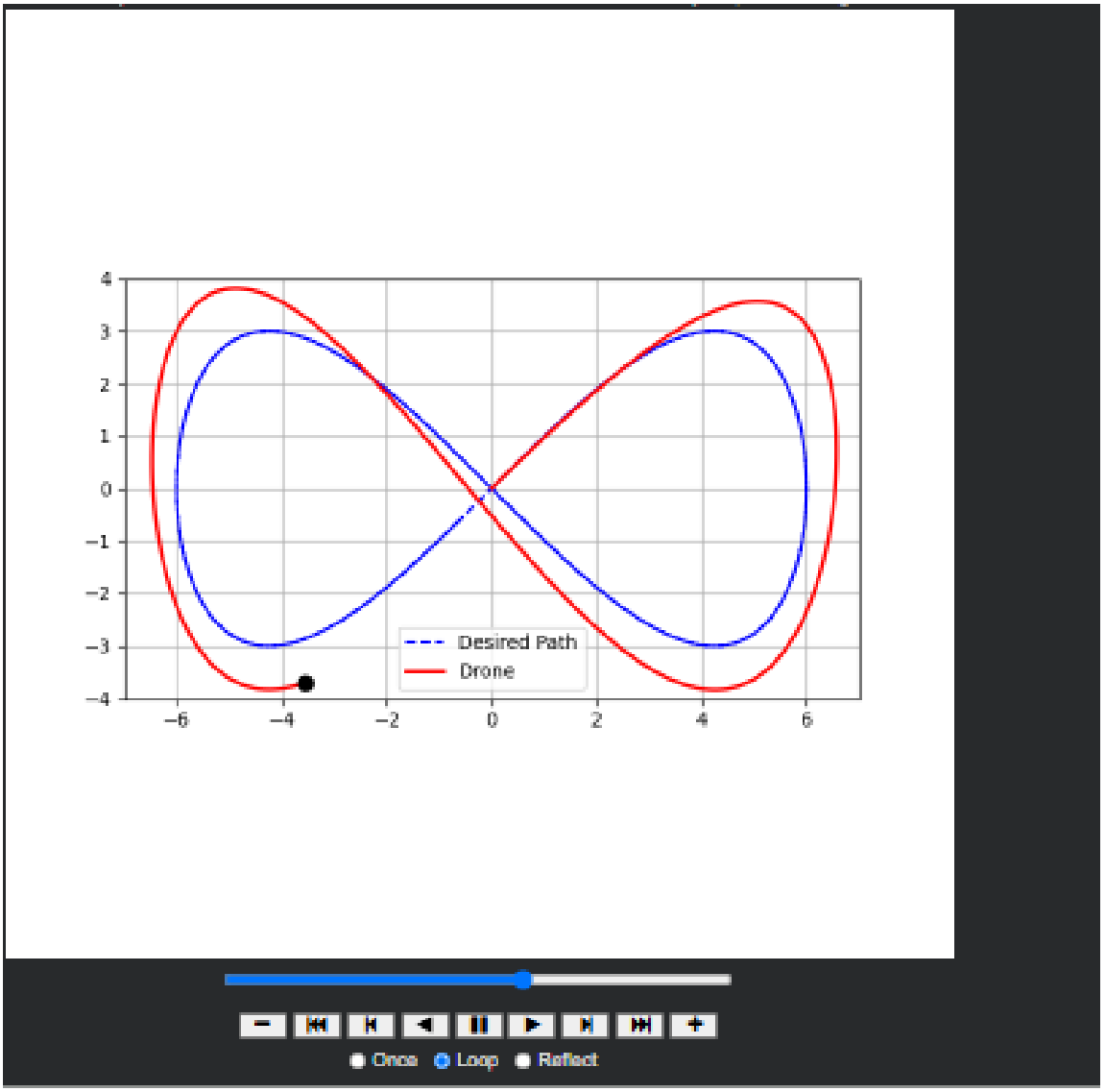


- Boat Animation



- Figure-Eight Simulation





Observations

- The PID controller successfully stabilized the drone altitude.
- Proper tuning of controller gains reduced oscillations and overshoot.
- Wind disturbance temporarily affected system performance, after which the controller restored stability.
- The autonomous boat accurately followed the desired path after tuning the controller parameters.
- Tracking accuracy was higher in the absence of environmental disturbances.

Conclusion

The project successfully demonstrated the practical implementation of guidance and control systems using Python simulations. PID controller tuning significantly improved the stability and disturbance rejection capability of the drone altitude control system. The autonomous boat guidance simulation demonstrated effective trajectory tracking under varying environmental conditions. Overall, the project provided valuable practical knowledge of feedback control, controller tuning, and autonomous navigation techniques used in aerospace engineering and robotics.

References

1. India Space Lab – Advanced Drone Technology Project Guidelines.
2. Python Documentation.
3. Google Colab Documentation.
4. NumPy Documentation.
5. Matplotlib Documentation.